

Toronto City Bylaw Conversational Agent

RSM8430 Final Project Proposal

TORONTO'S municipal bylaws and services are often complex and difficult for residents to navigate. We propose a conversational agent that combines RAG for accurate regulatory guidance with direct integration into 311 service workflows. Given over 100 structured bylaw chapters and queries that naturally split into information or action, this project supports the city's goal to modernize 311 services.

Knowledge Base

The system's knowledge base draws primarily from the [City of Toronto Municipal Code](#) and the [Toronto 311 Knowledge Base](#), with initial coverage focused on Noise, Animals, Waste Management, and Building Permits. The code's chapter-and-section organization provides naturally structured legal text that supports effective segmentation and retrieval in a RAG pipeline. All content is publicly available under the City of Toronto Open Government Licence, and the framework is designed to accommodate comparable open bylaw datasets from other jurisdictions such as Vancouver or New York City if needed.

Agent Capabilities

We implement three core actions to support resident needs. First, a **Hazard Reporter** collects hazard type, location, and severity through multi-turn dialogue. For example, a user reports a pothole and the agent requests location and severity before generating Ticket #TOR-HZ-1024.

Example Multi-Turn Flow

User: "I want to report a hazard on my street."
Agent: "What type of hazard? A pothole, fallen tree, or something else?"
User: "A big pothole."
Agent: "What is your postal code?"
User: "M5S 3E6"
Agent: "How severe is it — minor, moderate, or severe?"
User: "Severe, it's blocking a lane."
Agent: "Ticket #TOR-HZ-1024 submitted for a severe pothole near M5S 3E6."

Second, a **Permit Screener** helps residents determine

whether a home project requires a city permit. The agent collects project type and dimensions through multi-turn dialogue, then returns an eligibility decision with relevant bylaw citations. For example, a user asks whether they need a permit to build a backyard shed, the agent asks for the planned size, and if it is under 15 m², the agent confirms no permit is required. Third, a **Collection Lookup** lets residents check their local waste and recycling pickup schedule. The user provides a postal code and the agent returns the corresponding collection days and item categories. The system also includes emergency hand-offs for fire or crime, a legal advice disclaimer, illicit activity detection, and prompt injection defense.

Technical Approach

Implemented in Python, the system routes user queries through an intent classifier to either RAG retrieval, an action handler, or an out-of-scope response. The RAG pipeline retrieves relevant bylaw chunks from a vector store before the LLM generates a response with source attribution.

Technology Stack

LLM	qwen3-30b-a3b-fp8
Framework	LangGraph
Vector DB	ChromaDB
Embeddings	sentence-transformers
Frontend	Streamlit
State	SQLite

Evaluation will cover 12–15 test cases across RAG accuracy, intent routing, action completion, out-of-scope rejection, emergency hand-off, and prompt injection resistance.

Conclusion

This project addresses a clear civic need with a focused technical design. By separating knowledge retrieval from service actions and implementing robust guardrails, we ensure both accuracy and safety. Our evaluation plan confirms the system is ready for demonstration and aligns with Toronto's 311 modernization goals.